Mathematical Formulation of the Full-Horizon Middle-Mile Line-Haul Network Optimization Problem

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1. Problem Overview & System Representation

The middle-mile line-haul freight network spans a 31-day temporal horizon across 552 Origin-Destination (OD) candidate corridors and multiple sorting hubs. The primary operational objective is to determine a single, optimal, monthly dedicated daily fleet profile across all vehicle classes, assign cost-optimal direct or multi-hop hub-and-spoke transit paths for each demand lane, and selectively allocate at most 12% volumetric and physical freight spillover to third-party logistics (3PL) partners.

This document details the exact mathematical formulation of the champion monolithic Constraint Integer Programming (CIP) model implemented using Google OR-Tools CP-SAT.

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2. Sets and Indices

- N: Set of network sorting nodes and operational hubs, indexed by $i, j, h \in N$.

- T: Set of discrete operational days comprising the scheduling horizon, indexed by $t \in T = \{1, 2, \ldots, 31\}$.
- K: Set of available vehicle chassis types, indexed by $k \in K = \{1, 2, \ldots, 9\}$.
- $\mathcal{OD}$: Set of active freight demand lanes requiring transportation service, indexed by $od = (o, d) \in \mathcal{OD}$, where $o \neq d \in N$.
- P_{od}: Set of candidate transit routes pre-screened for lane od, indexed by $p \in P_{od}$. Each candidate path p is defined as an ordered sequence of physical arcs $A_p \subseteq A$ and intermediate transshipment hubs $H_p \subset N$.
- A: Set of active directed physical arcs traversing between facility pairs, $A = \bigcup_{od \in \mathcal{OD}} \bigcup_{p \in P_{od}} A_p$, indexed by $a = (i, j) \in A$.

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3. Deterministic Parameters

3.1. Demand & Service Requirements

- $W^{\text{phy}}_{od,t}$: Total physical demand weight (in kg) on lane $od \in \mathcal{OD}$ on day $t \in T$.
- $W^{\text{vol}}_{od,t}$: Total volumetric dimensional weight (in kg) on lane $od \in \mathcal{OD}$ on day $t \in T$.
- $W^{\text{phy}}_{\text{total}}$: Aggregate physical freight demand over the complete network horizon, $W^{\text{phy}}_{\text{total}} = \sum_{od \in \mathcal{OD}} \sum_{t \in T} W^{\text{phy}}_{od,t}$.
- $W^{\text{vol}}_{\text{total}}$: Aggregate volumetric freight demand over the complete network horizon, $W^{\text{vol}}_{\text{total}} = \sum_{od \in \mathcal{OD}} \sum_{t \in T} W^{\text{vol}}_{od,t}$.
- α^{spill}: Upper bound on the allowable physical and volumetric spillover fraction ($\alpha^{\text{spill}} = 0.12$).

3.2. Physical Route & Handling Characteristics

- D_a: Direct physical road distance (in km) along directed leg $a = (i, j) \in A$.
- D^{direct}_{od}: Direct highway distance from source node o to destination node d.
- δ_{od}: Allowable detour ceiling factor defined for lane od (typically $\delta_{od} = 1.50$).
- ω_a: One-way lane cost adjustment multiplier for leg $a \in A$ ($\omega_a \in [0.5, 1.0]$).
- c^{hop}_h: Loader handling cost per kg at sorting center $h \in N$ (derived from loader CPK data).

- γ: Soft quality regularizer penalty per kg applied to transshipment hops (γ = 0.20 Rs/kg).

3.3. Fleet Specifications

- Q_k^{phy}: Maximum physical payload capacity (in kg) of vehicle class $k \in K$.
- Q_k^{vol}: Maximum volumetric dimensional capacity (in kg) of vehicle class $k \in K$.
- F_k: Fixed vehicle dispatch/lease fee (Rs) incurred per round trip for chassis type $k \in K$.
- C_k^{km}: Variable running rate (Rs per km) for chassis class $k \in K$.
- $D_k^{\max}$: Maximum round-trip operational range limit (in km) for vehicle class $k \in K$.
- λ_k: Fleet regularization holding factor for dedicated chassis type $k \in K$ ($\lambda_k = 0.10 \times F_k \times |T|$).

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4. Decision Variables

4.1. Binary Strategic Network Variables

- $\theta_{od,p} \in \{0, 1\}$: Binary path selector, equal to 1 if lane $od \in \mathcal{OD}$ is routed along candidate route $p \in P_{od}$, and 0 otherwise.
- $s_{od} \in \{0, 1\}$: Binary spillover indicator, equal to 1 if lane $od \in \mathcal{OD}$ is diverted to third-party logistics (3PL) spillover, and 0 if served internally.

4.2. Integer Dispatch & Fleet Sizing Variables

- $y_{k,a,t} \in \mathbb{Z}_{\geq 0}$: Number of vehicle trips of chassis type $k \in K$ dispatched along physical directed arc $a \in A$ on date $t \in T$.
- $m_k \in \mathbb{Z}_{\geq 0}$: Constant committed daily fleet size for chassis class $k \in K$, enforcing the maximum dedicated units maintained across any day of the 31-day horizon.

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5. Objective Function

The objective function minimizes total operating line-haul dispatch expenses, intermediate hub handling transshipment costs, service quality penalties, and dedicated fleet capital holding regularizers across the entire 31-day planning horizon:

$$\min \ Z = Z_{\text{linehaul}} + Z_{\text{transshipment}} + Z_{\text{fleet_reg}} \tag{1}$$

Where:

$$Z_{\text{linehaul}} = \sum_{t \in T} \sum_{a \in A} \sum_{k \in K} \left(F_k + C_k^{\text{km}} \cdot D_a \right) \cdot \omega_a \cdot y_{k,a,t} \tag{2}$$

$$Z_{\text{transshipment}} = \sum_{od \in \mathcal{OD}} \sum_{p \in P_{od}} \theta_{od,p} \cdot \left[\sum_{t \in T} W_{od,t}^{\text{phy}} \cdot \left(\sum_{h \in H_p} c_h^{\text{hop}} + \gamma \cdot |H_p| \right) \right] \tag{3}$$

$$Z_{\text{fleet_reg}} = \sum_{k \in K} \lambda_k \cdot m_k \tag{4}$$

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6. System Constraints

6.1. Flow Conservation & Path Selection (Constraint C6)

Every lane $od \in \mathcal{OD}$ must either be assigned to exactly one candidate path or designated as 3PL market spillover:

$$\sum_{p \in P_{od}} \theta_{od,p} + s_{od} = 1 \quad \forall od \in \mathcal{OD} \tag{5}$$

6.2. Network Spillover Limits (Constraint C9)

The aggregate physical and volumetric demand allocated to 3PL spillover across the complete 31-day horizon cannot exceed the 12% commercial capacity cap:

$$\sum_{od \in \mathcal{OD}} s_{od} \cdot \left(\sum_{t \in T} W_{od,t}^{\text{phy}} \right) \leq \lfloor \alpha^{\text{spill}} \cdot W_{\text{total}}^{\text{phy}} \rfloor \tag{6}$$

$$\sum_{od \in \mathcal{OD}} s_{od} \cdot \left(\sum_{t \in T} W_{od,t}^{\text{vol}} \right) \leq \lfloor \alpha^{\text{spill}} \cdot W_{\text{total}}^{\text{vol}} \rfloor \tag{7}$$

6.3. Physical and Volumetric Leg Capacity Packing (Constraint C1)

For every active physical arc $a \in A$ and each day $t \in T$, the aggregate physical payload and volumetric dimensional capacity of the dispatched vehicle fleet must strictly cover the sum of demands traversing arc a:

$$\sum_{k \in K} Q_k^{\text{phy}} \cdot y_{k,a,t} \geq \sum_{od \in \mathcal{OD}} \sum_{\substack{p \in P_{od} \\ a \in A_p}} W_{od,t}^{\text{phy}} \cdot \theta_{od,p} \quad \forall a \in A,\ \forall t \in T \tag{8}$$

$$\sum_{k \in K} Q_k^{\text{vol}} \cdot y_{k,a,t} \geq \sum_{od \in \mathcal{OD}} \sum_{\substack{p \in P_{od} \\ a \in A_p}} W_{od,t}^{\text{vol}} \cdot \theta_{od,p} \quad \forall a \in A,\ \forall t \in T \tag{9}$$

6.4. Constant Daily Fleet Sizing Commitment (Constraint C7)

The daily deployment of chassis type k across the entire physical network cannot exceed the contractual committed constant daily fleet envelope m_k:

$$\sum_{a \in A} y_{k,a,t} \leq m_k \quad \forall k \in K,\ \forall t \in T \tag{10}$$

6.5. Path Feasibility & Operational Bounds (Section 5 / C8, S1, S2)

By construction of the candidate path generation set P_{od}, every allowable route candidate strictly conforms to the following operational parameters:

$$\text{Detour Ceiling (C8):} \quad \sum_{a \in A_p} D_a \leq \delta_{od} \cdot D_{od}^{\text{direct}} \quad \forall p \in P_{od},\ \forall od \in \mathcal{OD} \tag{11}$$

$$\text{Max Hops (S1):} \quad |H_p| \leq 2 \quad \forall p \in P_{od},\ \forall od \in \mathcal{OD} \tag{12}$$

$$\text{Chassis Range Limit:} \quad 2 \cdot D_a \leq D_k^{\max} \quad \forall a \in A,\ \forall k \in K \text{ where } y_{k,a,t} > 0 \tag{13}$$

6.6. Non-Negativity and Integrality Restrictions

$$\theta_{od,p} \in \{0, 1\} \quad \forall od \in \mathcal{OD},\ \forall p \in P_{od} \tag{14}$$

$$s_{od} \in \{0, 1\} \quad \forall od \in \mathcal{OD} \tag{15}$$

$$y_{k,a,t} \in \{0, 1, 2, \ldots, 60\} \quad \forall k \in K,\ \forall a \in A,\ \forall t \in T \tag{16}$$

$$m_k \in \{0, 1, 2, \ldots, 350\} \quad \forall k \in K \tag{17}$$

7. Heuristic Reformulations and Constraint Transformations

While the exact formulation in Section 6 models the complete polytope simultaneously, computational intractability and memory ceilings require heuristic approximations that alter, relax, or decompose specific constraint sets:

7.1. Rolling-Horizon Matheuristic (Temporal Decomposition)

- **Decomposed Set:** The full time horizon T is partitioned into overlapping windows $T_w \subset T$ of size $|T_w| = 7$ with step $\Delta = 5$.
- **Transformed Constraint (C7):** The strict simultaneous coupling $\sum_{a \in A} y_{k,a,t} \leq m_k$ is relaxed. Dedicated fleet stability across temporal boundaries is enforced via penalty terms in the local objective:

$$\min \; Z_w + \sum_{k \in K} \mu_k \cdot \left| m_k^{(w)} - \bar{m}_k \right| \tag{18}$$

  where $\bar{m}_k$ is the anchored fleet size from window $w - 1$.

7.2. Chassis-Centric ALNS (Metaheuristic Abstraction)

- **Vanished Constraint Matrix (C1):** The discrete multi-class vehicle integer packing inequalities $\sum_k Q_k y_{k,a,t} \geq \dots$ are removed from the solver matrix.
- **Surrogate Evaluation:** Vehicle requirements are approximated using a continuous upper envelope dominated by the primary 32 FT multi-axle chassis:

$$Y_{a,t} = \max \left(\left\lceil \frac{F_{a,t}^{\text{phy}}}{Q_{32\text{FT}}^{\text{phy}}} \right\rceil , \; \left\lceil \frac{F_{a,t}^{\text{vol}}}{Q_{32\text{FT}}^{\text{vol}}} \right\rceil \right) \tag{19}$$

  The combinatorial trade-offs among the remaining 8 chassis classes vanish during neighborhood search.

7.3. Fix-and-Optimize Matheuristic (Dimensional Projection)

- **Restricted Decision Space:** In iteration round r, lane set $\mathcal{OD}$ is partitioned into an active subset $\mathcal{OD}_{\text{free}}^{(r)}$ and a frozen subset $\mathcal{OD}_{\text{fixed}}^{(r)}$.
- **Constraint Fixing:** For all $od \in \mathcal{OD}_{\text{fixed}}^{(r)}$, routing variables $\theta_{od,p}$ and spillover variables s_{od} are fixed to their incumbent values $\theta_{od,p}^*$ and s_{od}^*, projecting the constraint matrix onto a reduced dimensional subspace.

7.4. Evolutionary Genetic Algorithm (Unconstrained Penalty Formulation)

- **Vanished Constraints (C6, C9):** Hard algebraic linear inequalities are omitted. Constraint satisfaction is maintained via chromosomal gene bounds (for C6) and exterior penalty functions in the fitness objective (for C9):

$$\text{Fitness}(\boldsymbol{x}) = Z_{\text{eval}}(\boldsymbol{x}) + M \cdot \max\left(0,\ \text{Spillover}(\boldsymbol{x}) - \alpha^{\text{spill}}\right) \tag{20}$$

  where $M = 10^{12}$ serves as a barrier multiplier against infeasible spillover allocations.